

Future Enhancements

1. Purpose

The current version of the Expense Tracker application provides the core functionality for personal finance management. In future releases, additional features will be introduced to improve usability, automation, security, and financial insights.

2. Current Limitations

The current version of the application has some limitations because development is currently focused on the frontend and UI implementation.

Frontend limitations

- Some screens currently use static/local data.
- Financial data is not yet persisted to a backend database.
- Some menu options are currently UI-only.
- Payment methods are represented as UI screens and are not connected to real payment services.
- Wallet functionality is currently a frontend implementation.
- Some functionality will depend on the future backend implementation.

Backend limitations

The Django backend has not yet been implemented.

Therefore, the following are currently pending:

- User authentication through the backend
- Database persistence
- Transaction APIs
- Wallet APIs
- Profile APIs
- Authentication/token management
- Server-side validation

- Backend error handling
 - Secure financial data management
-

3. Short-Term Enhancements (Version 1.1)

The following features are planned immediately after completing the current application.

User Profile Management

- View Profile
 - Edit profile information.
 - Upload profile picture.
 - Change password.
 - Profile Navigation
 - Profile Menu
 - Update email address.
 - Log out
-

Transaction Management

- Edit existing transactions.
 - Delete transactions.
 - Search transactions.
 - Filter by category.
 - Filter by date.
 - Sort by amount.
 - Sort by newest or oldest.
-

Statistics Improvements

- Monthly analytics.

- Weekly analytics.
 - Yearly analytics.
 - Interactive charts.
 - Pie charts.
 - Bar charts.
 - Line charts.
-

Wallet Improvements

- Add multiple wallets.
 - Add multiple bank accounts.
 - Wallet transfer.
 - Card management.
 - Wallet balance synchronization.
-

4. Medium-Term Enhancements (Version 2.0)

Budget Planning

Users will be able to:

- Create monthly budgets.
 - Create weekly budgets.
 - Receive budget alerts.
 - View remaining budget.
 - Budget progress indicator.
-

Savings Goals

Users will be able to:

- Create savings goals.

- Set target amounts.
- Track progress.
- Receive milestone notifications.

Example:

- Buy Laptop
 - Vacation Fund
 - Emergency Fund
 - Car Savings
-

Recurring Transactions

Support automatic recurring transactions.

Examples:

- Salary
 - House Rent
 - Electricity Bills
 - Netflix Subscription
 - Spotify Subscription
-

Bill Reminder System

Users will receive reminders before bill due dates.

Examples:

- Electricity Bill
 - Gas Bill
 - Water Bill
 - Internet Bill
-

Notification System

Push notifications for:

- Upcoming bills.
 - Monthly reports.
 - Budget alerts.
 - Savings milestones.
 - Transaction confirmations.
-

5. Long-Term Enhancements (Version 3.0)

AI Expense Analysis

Artificial Intelligence will:

- Analyze spending habits.
 - Detect unusual expenses.
 - Suggest savings opportunities.
 - Recommend budget improvements.
 - Predict future expenses.
-

Receipt Scanner

Users will be able to:

- Scan receipts.
- Automatically extract:
 - Store name
 - Date
 - Amount
 - Category

OCR Integration

Automatically convert receipt images into transactions.

Voice Assistant

Users can add expenses using voice commands.

Examples:

"Add a food expense of \$20."

"Record salary of \$500."

Multi-Currency Support

Support:

- USD
- PKR
- EUR
- GBP
- SAR
- AED

Automatic exchange rate updates will be provided.

Dark Mode

Users can switch between:

- Light Theme
 - Dark Theme
 - System Theme
-

Cloud Synchronization

Synchronize data across multiple devices.

Examples:

- Mobile
 - Tablet
 - Web Application
-

Export Reports

Export reports as:

- PDF
 - Excel
 - CSV
-

Biometric Authentication

Support:

- Fingerprint Login
 - Face ID
 - Device PIN Authentication
-

Offline Mode

Allow users to:

- Add transactions without internet.
 - Synchronize automatically when the connection is restored.
-

Web Version

Develop a web version using React.js with the same backend.

6. Django Backend Enhancements

Once the Django backend is integrated, the following features will be available:

- JWT Authentication
 - Secure REST APIs
 - PostgreSQL Database
 - Cloud File Storage
 - Email Verification
 - Password Recovery
 - Role-Based Access Control
 - API Rate Limiting
 - Activity Logging
-

7. Security Improvements

Future versions will include:

- Two-Factor Authentication (2FA)
 - End-to-End Encryption
 - Secure API Authentication
 - SQL Injection Protection
 - Brute Force Protection
 - Automatic Session Timeout
 - Secure Password Policies
-

8. Performance Improvements

Future optimizations include:

- Faster application startup.
- Optimized image loading.
- Reduced memory usage.
- Improved API response times.
- Lazy loading of screens.
- Code splitting.
- Local caching for frequently accessed data.

9. Project Roadmap

Version	Planned Features
---------	------------------

Version 1.0	Authentication, Dashboard, Statistics, Wallet, Connect Wallet
--------------------	---

Version 1.1	Profile Management, Transaction Management, Interactive Charts
--------------------	--

Version 2.0	Budget Planning, Savings Goals, Notifications, Bill Reminders
--------------------	---

Version 3.0	AI Insights, OCR, Voice Commands, Multi-Currency, Cloud Sync
--------------------	--

Version 4.0	Web Application, Advanced Analytics, Investment Tracking
--------------------	--